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Unit – 5 : Session – 9 : SLO – 1 | Instruction Formats (Zero/One/Two/Three-Address)

■ I. Multiple Choice

1. C) Stack — zero-address instructions operate on the stack.
2. D) Accumulator — the one-address format uses the accumulator as the implicit operand.
3. B) Destination operand — in the two-address format the result overwrites the destination.

■ II. Fill in the Blanks

1. postfix — zero-address format is ideal for postfix (reverse-Polish) expressions.
2. accumulator — one-address format uses the accumulator for intermediate calculations.
3. three — the three-address format explicitly specifies three operands (two sources + destination).

■ III. True or False

1. FALSE — zero-address instructions take operands implicitly from the stack.
2. TRUE.
3. TRUE — an explicit destination removes the need for temporary memory.

■ IV. Match the Following

- A) Zero Address — 1) Stack-based.
B) One Address — 2) Accumulator.
C) Two Address — 3) Destination overwritten.
D) Three Address — 4) Three explicit operands.

■ V. Numerical Answers

1. Zero-address code for $(A + B) * (C + D)$: PUSH A; PUSH B; ADD; PUSH C; PUSH D; ADD; MUL — the stack machine leaves $(A+B)*(C+D)$ on top.
2. 4 bits — $\log_2(16) = 4$ bits encode 16 registers.
3. 24 bits — 8-bit opcode + 16-bit address = 24 bits per instruction.